MTH219 2023 S2 TOA

Feel free to contact me at maths_and_programming@outlook.sg if you need tuition. I provide university level tuition for the following SUSS units:

* ICT133 Structured Programming
* ICT162 Structured Programming
* MTD215 Application of C++ in Multimedia
* ICT231 Data Structures and Algorithms I
* ICT232 Data Structures and Algorithms II
* MTH105 Fundamentals of Mathematics
* MTH107 Calculus I
* MTH108 Calculus II
* MTH316 Multivariable Calculus
* MTH207 Linear Algebra
* MTH219 Fundamentals of Statistics and Probability
* MTH220 Statistical Methods and Inference

Donaldson Tan

Q1 — 430 pm
Q2 — 430
Q3 — 500
Q4 — 530
end — 600

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SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

MTH219

Examination - July Semester 2023

Fundamentals of Statistics and Probability

Thursday, 16 November 2023 **04:00 pm - 06:00 pm**

12 + 3 = 15

Time allowed: 2 hours

5, 4, 1, 2, 3

70

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FOUR (4)** questions and comprises **FIVE (5)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.
4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1a(i)

A random variable X follows a normal distribution with mean μ and variance σ^2, where $\sigma = 30$. Given that $P(X \leq 25.73) = 0.70$, determine the mean μ.

(4 marks)

Question 1a(ii)

Draw a sketch of the PDF of X, clearly indicating the mean, median, 1σ from mean, 2σ from mean.

(4 marks)

Question 1a(iii)

Compute the probability that a random sample of X lands within 1σ of the mean.

(3 marks)

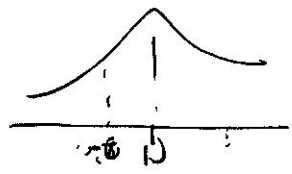
Question 1a(iv)

Let the variable T be the total value of 5 samples of X that follows the normal distribution given in Question 1a(i), i.e. $T = X_1 + X_2 + X_3 + X_4 + X_5$. Compute $P(T \leq 0)$.

(4 marks)

Question 1b

A continuous random variable X has distribution $N(\mu, \sigma^2)$. It is known that $P(X < a) = 0.4$ and $P(X < b) = 0.7$.



Question 1b(i)

Sketch a normal curve with mean μ and variance σ^2. On the graph, mark out the regions $\{X \in \mathbb{R} | \, P(X < a)\}$ and $\{X \in \mathbb{R} | \, P(X \geq b)\}$ where a and b are real numbers such that $P(X < a) = 0.4$ and $P(X < b) = 0.7$.

(5 marks)

Question 1b(ii)

Given that $\mu = 0$ and $\sigma^2 = 1$, calculate the value of a and b.

(5 marks)

Question 2a

A sample of 20 readings were taken from an experiment and the results are shown below:

19.8	8.8	2.2	4.4	11
5.6	17.7	6.6	4.4	8.8
8.8	8.8	8.8	13.2	8.8
17.7	19.8	17.7	2.2	17.6

Table Q2(a)

Analyse and calculate the following sample descriptive statistics:

Mean, median, mode, range, standard deviation, variance, lower quartile, upper quartile, interquartile range, 10% percentile, coefficient of variation

Show your working, a short description of what the statistic means, formula, and assumptions clearly. Answers directly computed from software/calculator without working will not be granted marks.

(15 marks)

Question 2b

A data analyst thought that the data appears to resemble a Uniform (0,22) distribution. Compute the Mean, median, mode, range, standard deviation, variance, interquartile range, coefficient of variation of the U(0,22) distribution and make a comparison between U(0,22) and the dataset in Question 2a. Explain your reasoning clearly.

Hint: For U(0,22), the lower quartile is 5.5, Upper quartile is 16.5, 10th percentile is 2.2

(10 marks)

$P(\text{pair}) = \sum_{i=1}^{8} P(X_1=i)\, P(X_2=i) = \frac{1}{8}$

$P(\text{rnb}) = 2\sum_{i=3}^{8} \sum_{j=1}^{8} P(X_i = i)\, P(X_2=j) = \frac{21}{32}$

Question 3a

In a dice game, two fair 8-sided dice (Octahedron) are rolled and prices are given based on the different combinations of the rolls. The 8-sided dice are numbered from 1 to 8.

Here are the rules of the game:

Rule 1: If a pair, i.e. (1,1), (2,2), etc are rolled, the banker loses \$1 and the player wins \$1.
Rule 2: If the combined roll is equal to or above 10, the banker loses \$1 and the player wins \$1.
Rule 3: Otherwise, the banker will win \$1, the player loses \$1.

Question 3a(i) P(

Prove that the game is fair, i.e. the expected winnings are \$0 for either party.

(7 marks)

Question 3a(ii)

The banker would like to change the payouts to make the game more interesting. The same rules from Question 3(a) apply but the winnings are now different.

- The player wins Double of the sum of the rolls if it is a pair (i.e. If the player rolls 8-8, he will win 16*2 = 32.)
- The player wins Equal to the sum of the rolls if it is greater than or equal to 10, except when it is a pair.
- Otherwise, the player loses \$$X$.

Calculate the value of X, such that the banker makes an average of \$1 per game.

(8 marks)

Question 3b

A cake factory receives egg shipments from 2 different suppliers.

$-x \quad -y$
$+(1-x-y)$
$-2x \quad -2y+1$

Supplier A sends 2000 eggs a day, with 0.05% of eggs arriving broken.
Supplier B sends 3000 eggs a day, with 0.07% of eggs arriving broken.

Let X and Y be the number of broken eggs from Supplier A and B on a random day respectively. State the distribution, mean and variance of X and Y.

(4 marks)

Question 3c

Using the Poisson approximation or otherwise, compute the probability of $P(X+Y>4)$.

(6 marks)

$(1 - P(X=0))$

$\binom{5}{1} p(1-p)^2 \binom{6}{2} p^2 (1-p)^4 \ldots$

Question 4

A group of tourists went to an African Safari to watch the wildlife. In the area that they visited, they are likely to see Lions, Rhinos, Giraffes and other animals.

Based on the experience of the guide, the guide estimates that in 3 hours, the average number of sightings for Lions, Rhinos, Giraffes and other animals are 3, 5, 4 and 2 respectively. We assume that a Poisson model is adequate to describe the number of sightings of each animal.

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Question 4a

Compute the expected value and variance of the number of animals sighted in 3 hours.

(5 marks)

Question 4b

Apply the Poisson model to determine the probability that the guests will not see a single animal in 3 hours.

(3 marks)

Question 4c

To prevent disappointment to the guests, if less than 3 lions were sighted in the first 3 hours, the guide will work overtime and offer to bring the guests in the safari for another 1.5 hours. What is the probability that the tourists will not be able to see any Lions in a single day?

$3+3+3+1.5$ (5 marks)

Question 4d

Assuming that the guide has a 5 day work week, what is the probability that he will work for more than 16.5 hours in that week given that more than 3 lions were sighted in the first 3 hours on Monday and Tuesday?

W.5.

(7 marks)

Question 4e

Assuming that today is Monday and the guide did overtime today, calculate the probability that his next overtime is done on Friday, assuming he works on all weekdays.

(5 marks)

----- END OF PAPER -----

1(a) i. $X \sim N(\mu, \sigma^2)$ where $\sigma = 30$

$P(X \leq 25.73) = 0.7$

$$\frac{25.73 - \mu}{\sigma} = \text{qnorm}(0.7)$$

$\therefore \mu = 25.73 - \sigma \times \text{qnorm}(0.7)$

$= 9.999985$

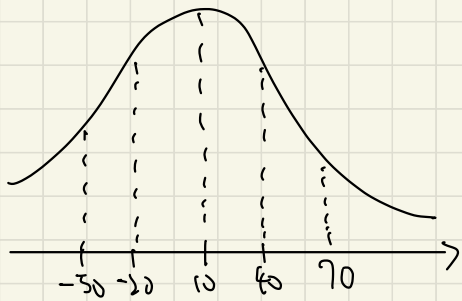
$\simeq 10$

ii. $\mu - \sigma = -20.00202 \simeq -20$

$\mu + \sigma = 39.99798 \simeq 40$

$\mu - 2\sigma = -50.00202 \simeq -50$

$\mu + 2\sigma = 69.99798 \simeq 70$



iii required probability $= P(-1 \leq Z \leq 1)$

$= \text{pnorm}(1) - \text{pnorm}(-1)$

$= 0.6826895$

$\simeq 0.683$

iv. $T = \sum_{i=1}^{5} X_i$

$E(T) = E\left(\sum_{i=1}^{5} X_i\right) = 5E(X) = 5\mu$

$\text{var}(T) = \text{var}\left(\sum_{i=1}^{5} X_i\right) = 5\,\text{var}(X) = 5\sigma^2$

$T \sim N(5\mu, 5\sigma^2)$

$P(T \leq 0) = \text{pnorm}(0, 5\mu, \sqrt{5\sigma^2})$

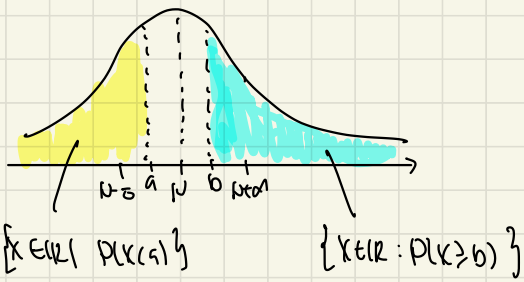
$= 0.228$

1(b) i. $P(X < \mu - \sigma) = \text{pnorm}(-1) = 0.1586553$

$P(X < \mu) = 0.5$

$P(X < \mu + \sigma) = \text{pnorm}(1) = 0.8413447$

$\therefore \mu - \sigma < a < \mu$ and $\mu < b < \mu + \sigma$



ii. $a = \text{qnorm}(0.4, 0, 1)$

$= -0.2533471 \simeq -0.25$

$b = \text{qnorm}(0.7, 0, 1)$

$= 0.5244005 \simeq 0.52$

2(a) $X = (2.2, 2.2, 4.4, 4.4, 5.6,$
$6.6, 6.6, 6.6, 6.6, 6.6,$
$6.6, 9.9, 11, 13.2, 17.6,$
$17.7, 17.7, 17.7, 19.8, 19.8)$

$\min(X) = 2.2$, $\max(X) = 19.8$

range $= \max - \min = 17.6$

mean $= \frac{1}{20}\sum_{i=1}^{20} x_i = 10.69$

variance $= \frac{1}{20-1}\sum_{i=1}^{20}(x_i - \bar{x})^2 = 34.31$

standard deviation $= \sqrt{\text{variance}} = 5.86$

coefficient of variation $= \frac{\sigma}{\bar{x}} = \frac{5.86}{10.69} = 54.8\%$

lower quartile $= X_{0.25(n+1)} = X_{5.25} = X_5 + 0.25(X_6 - X_5) = 5.6 + 0.25(6.6 - 5.6) = 5.85$

upper quartile $= X_{0.75(n+1)} = X_{15.75} = X_{15} + 0.75(X_{16} - X_{15}) = 17.6 + 0.75(17.7 - 17.6) = 17.675$

interquartile range $= X_{15.75} - X_{5.25} = 11.825$

10th percentile $= X_{0.10(n+1)}$
$= X_{2.1}$
$= X_2 + 0.1(X_3 - X_2)$
$= 2.2 + 0.1(4.4 - 2.2)$
$= 2.42$

2(b) $X \sim U[0, 22]$

mean $=$ median $= \frac{a+b}{2} = \frac{0+22}{2} = 11$

there is no mode

range $= \max(X) - \min(X) = 22$

variance $= \frac{(b-a)^2}{12} = \frac{22^2}{12}$

standard deviation $= \sqrt{\text{variance}} = \frac{22}{\sqrt{12}} = \frac{11}{\sqrt{3}}$

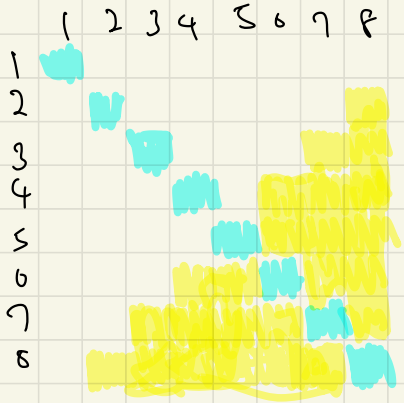
coefficient of variation $= \frac{\sigma}{\bar{x}} = \frac{11}{\sqrt{3}} \div 11 = \frac{1}{\sqrt{3}}$

interquartile range $=$ upper quartile $-$ lower quartile
$= 16.5 - 5.5$
$= 10$

3(a) i. X: A pair is obtained

Y: combined roll ≥ 10

Let D_1, D_2 be the dice roll of the 2 dices



$$P(X) = \frac{8}{64} = \frac{1}{8}$$

$$P(Y) = \frac{24}{64} = \frac{3}{8}$$

$$P(\text{otherwise}) = 1 - \frac{1}{8} - \frac{3}{8} = \frac{1}{2}$$

$$E(\text{player's winning}) = \$1 \times [P(X) + P(Y)] - \$1 \times P(\text{otherwise})$$

$$= \$1 \times \left[\frac{1}{8} + \frac{3}{8}\right] - \$1 \times \frac{1}{2}$$

$$= 0$$

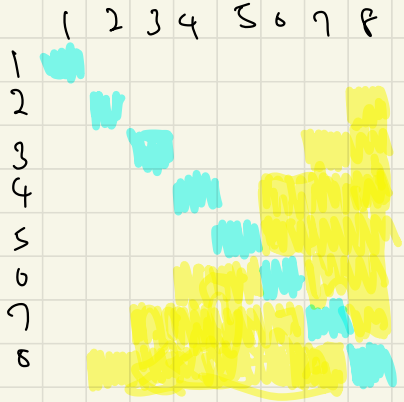
$$E(\text{banker's winning}) = -\$1\,[P(X) + P(Y)] + \$1 \times P(\text{otherwise})$$

$$= -\$1 \times \left[\frac{1}{8} + \frac{3}{8}\right] + \$1 \times \frac{1}{2}$$

$$= 0$$

$\therefore$ the game is fair

3(a) ii.



E(player's winnings from event X)

$$= \sum_{i=1}^{8} 4i \cdot P(D_1=i) \cdot P(D_2=i)$$

$$= \sum_{i=1}^{8} \frac{4}{64} i$$

$$= \$\frac{1}{16} \sum_{i=1}^{8} i$$

$$= \$\frac{1}{16} \cdot \frac{8(8+1)}{2}$$

$$= \$\frac{9}{4}$$

E(player's winning from Y)

$$= \$2 \times [3\times 0 + 3\times 4 + 2\times 12 + 2\times 13 + 14 + 15] \times \frac{1}{64}$$

$$= \$2 \times [30 + 33 + 24 + 26 + 14 + 15] \times \frac{1}{64}$$

$$= \$\frac{71}{16}$$

E(player's winning from otherwise)

$$= \$ - X \cdot P(\text{otherwise})$$

$$= \$ - \frac{X}{2}.$$

let E(player's winning) = − E(banker's winning) = −\$1

$$\frac{9}{4} + \frac{71}{16} - \frac{X}{2} = -1$$

$$X = 2\left(\frac{9}{4} + \frac{71}{16} + 1\right)$$

$$= \frac{123}{8}$$

3b $X \sim Bin(2000, 0.0005)$

$E(X) = np = 0.1$

$var(X) = npq = 0.099915$

$Y \sim Bin(3000, 0.0007)$

$E(Y) = np = 2.1$

$var(Y) = npq = 2.09853$

(c) let X_p be Poisson approximation of X

let Y_p be Poisson approximation of Y

$X_p \sim Po(0.1)$

$Y_p \sim Po(2.1)$

$X_p + Y_p \sim Po(2.2)$

$P(X+Y > 4) \simeq P(X_p + Y_p > 4)$

$= 1 - P(X_p + Y_p \leq 4)$

$= 1 - ppois(4, 2.2)$

$= 0.0725$

4(a) $L \sim Po(3)$, $R \sim Po(5)$, $G \sim Po(4)$ $O \sim Po(2)$

$T = L + R + G + O$

$T \sim Po(14)$

$E(T) = \lambda = 14$

$Var(T) = \lambda = 14$

(b) $P(T=0) = dpois(0, 14)$

$= 8.315 \times 10^{-7}$

(c) $L_{OT} \sim Po(1.5)$

$P(\text{tourists seen no lion}) = P(L=0) \times P(L_{OT}=0)$

$= dpois(0,3) \times dpois(0,1.5)$

$= 0.0111$

(d) Work more than 16.5h $\Rightarrow$ work > 1 OT shift

$p = P(L<3) = P(L \leq 2) = ppois(2,3)$

$P(>3 \text{ lions sighted per day on Mon, Tue}) = (1-p)^2$

$P(>1 \text{ OT} \cap 3 \text{ lions sighted per day on Mon, Tue}) = (1-p)^2 \binom{3}{2} p^2 (1-p) + (1-p^2)\binom{3}{3} p^0 (1-p)^3$

$= (1-p)^2 [3p^2(1-p) + (1-p)^3]$

$P(\text{work} > 16.5h \text{ in this week} \mid >3 \text{ lions sighted on Mon, Tue})$

$= \dfrac{(1-p)^2(3p^2(1-p) + (1-p)^3)}{(1-p)^2}$

$= 3p^2(1-p) + (1-p)^3$

$= 0.501826$

$\simeq 0.502.$

4(d) $P(\text{next OT is Friday}) = (1-p)^3 p$

$= 0.0821451$

$\simeq 0.0812$